Cytosim-based simulation to study neurite formation from symmetric Neuronal Progenitor Cells (NPCs)

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ABSTRACT

Neuronal polarisation and emergence of a single axon from an isotropic cell body is a fundamental symmetry-breaking event whose cytoskeletal basis remains incompletely understood. Dissecting the **minimal motor forces** sufficient to drive directed membrane protrusion requires quantitative simulation inaccessible to live-cell experiments alone. **Oelz et al. (2018)** showed that cortical dynein, kinesin-1, and dynamic microtubules cooperatively generate stable, localised protrusions resembling nascent neurites. Since Cytosim does not natively deform boundaries mid-simulation, we built a **Python orchestrator** that reads MT confinement forces, deforms the boundary polygon, and rewrites the simulation file iteratively to reproduce stable multi-process morphologies coupled with polarity sorting.

BACKGROUND

How do neurons establish polarity?

Early work by **Bradke & Dotti (1997)** showed that the axon-specified neurite contains a more dynamic, less stable actin network, implicating cytoskeletal asymmetry as a driver of neuronal polarity rather than a consequence of it.

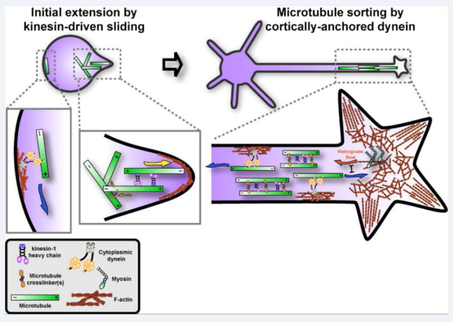
Witte, Neukirchen & Bradke (2008) demonstrated that microtubule (MT) dynamics are not uniform across neurites — the future axon has selectively stabilised MTs, establishing MT polarity sorting as a key step in axon specification.

Baas et al. (1988) established that axonal MTs are uniformly oriented (plus-end-out), while dendritic MTs are mixed-polarity. This polarity sorting underpins directed vesicle transport and is essential for proper neuronal function.

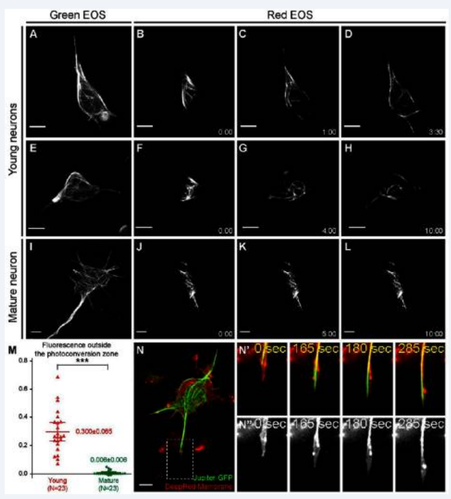
Kapitein & Hoogenraad (2011) unified these findings into a model where MT polarity is actively maintained by motor proteins: kinesin-1 slides antiparallel MTs apart, while cortical dynein captures and orients MT plus-ends outward — the exact mechanism modelled by **Oelz, del Castillo, Gelfand & Mogilner (2018)**.

Wen Lu & Gelfand (2018) further showed in *Drosophila* neurons that dynein-driven MT sliding from the cell body into nascent neurites is sufficient to initiate and sustain neurite formation even in the absence of directed actin polymerisation, directly motivating the cortical dynein placement rules implemented in this simulation.

Oelz, del Castillo, Gelfand & Mogilner (2018) synthesised this body of work into a full computational and mathematical model. Cortical dynein anchored at the cell periphery pulls MT plus-ends inward; kinesin-1 crosslinks and slides antiparallel MT pairs outward; dynamic MT growth pushes against the membrane. Together these three forces produce a stable, self-organised protrusion that steers by asymmetric dynein placement.



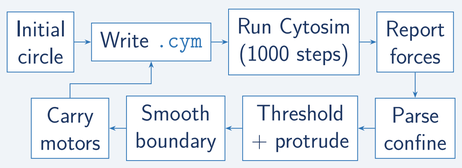
Wen Lu & Gelfand (2018)



Wen Lu & Gelfand (2014)

CYTOSIM-PYTHON LOOP ARCHITECTURE

The simulation couples Cytosim (a Langevin-dynamics fiber engine) with a Python boundary-deformation layer. Each epoch advances 1 s of simulated time:



- **MT positions** from `fiber:point` are re-injected as explicit shape blocks each epoch, preserving spatial history across epoch boundaries.
- **Dynein and kinesin** positions (from `dynein_cortex:position` and `kinesin1:state` reports) are parsed and re-injected.
- Polygon file updated each epoch; Cytosim reads the deformed boundary at the start of the next run.

PROTRUSION RULE — STAGES A TO E

Per-vertex threshold:

$$\theta_i = F_p + \kappa_{\mathrm{loc}}\, e_i + \kappa_{\mathrm{ten}} \sum_j e_j, \qquad e_i = \max(0,\ r_i - R_0)$$

Vertex i is active when $f_i > \theta_i$; displaced radially by Δr; neighbours get 50 % of displacement.

- **Stage A** — bin confine forces onto nearest vertex ($<1.5\,\mu$m); accept only when radial fraction $f_r/|\mathbf{F}| \geq 0.5$ (tangential forces discarded).
- **Stage B** — Gaussian smooth f_{seg} along boundary ring ($\sigma = 2.0$ vertices, wrapped).
- **Stage C** — compute per-vertex threshold (two classes: flat membrane $F_p = 3.0$ pN; process vertices). Global tension penalty grows with total extension.
- **Stage D** — protrude active vertices; hard cap `MAX_EXT_UM` $= 8\,\mu$m.
- **Stage E** — post-protrusion Gaussian filter on radial extension ($\sigma = 1.5$ vertices); eliminates single-vertex spikes.

MT COORDINATE CARRY-FORWARD

At epoch > 0, node positions $\{a_i\}$ from `fiber:point` are re-injected using the **local frame** convention:

$$s_i = a_i - a_0, \quad \texttt{position} = a_0 \implies r_i = s_i + a_0 = a_i$$

MT end classification: an end is counted only when (i) $r > R_0$ (outside cell body) and (ii) its angle falls within the process sector ($\pm\pi n_v / N_{\mathrm{seg}}$ of tip angle).

MOTOR INITIALIZATION

Kinesin-1 (`set couple kinesin1, activity = crosslink`)

- Motor hand walks toward MT **plus end** at $+0.57\,\mu$m/s; cargo hand anchors on a second MT.
- Slides antiparallel MT pairs, pushing minus ends outward toward the cortex.
- Stall force 4.7 pN; binding range $0.02\,\mu$m.
- *v4:* positions from `kinesin1:state` (columns 3–4) re-injected each epoch.

Cortical dynein (`set single dynein_cortex, activity = fixed`)

- Cargo pinned to cortex; motor walks toward MT **minus end** at $-0.86\,\mu$m/s.
- MT transported plus-end-forward toward cortex.
- Stall force 1.36 pN; binding reach $0.75\,\mu$m.
- *Code:* positions from `dynein_cortex:position` (columns 2–3) carried forward; new dyneins placed with process-biased weighting.

CYTOSIM: NUMERICAL FRAMEWORK (NÉDÉLEC & FOETHKE 2007)

Overdamped Langevin equation

$$\mathrm{d}\mathbf{x} = \mu\, F(\mathbf{x}, t)\,\mathrm{d}t + \mathrm{d}B(t)$$

Inertia ignored at cellular scale. μ = mobility; F = total force; $\mathrm{d}B$ = Brownian noise.

Implicit integration

$$[I - \tau P_t \mu A_t]\,\Delta\mathbf{x} = P_t[\tau\mu(A_t \mathbf{x}_t + G_t) + \delta B_t]$$

Unconditionally stable.

Fiber bending + constraint

$$F_{\mathrm{bend}} = \kappa\left(\frac{p}{L}\right)^3 (m_{k-1} - 2m_k + m_{k+1})$$
$$P = I - J^t(JJ^t)^{-1}J$$

$\kappa = 10\,\mathrm{pN}\cdot\mu\mathrm{m}^2$. P enforces constant segment length $|m_{k+1} - m_k| = L/p$.

Motor kinetics + confinement

$$\delta a = \tau v_{\max}\left(1 - \frac{f}{f_{\mathrm{stall}}}\right), \quad p_{\mathrm{off}} = p_0\, e^{|f|/f_0}$$
$$f(a) = k(p(a) - a), \ k = 200\,\mathrm{pN}/\mu\mathrm{m}$$

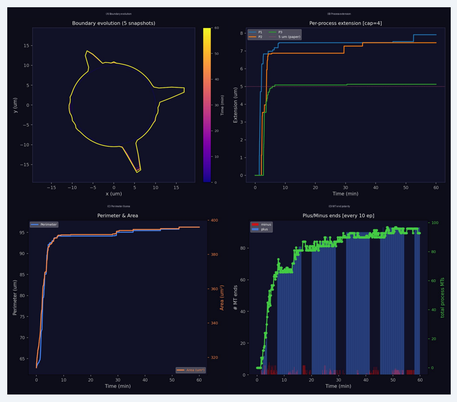
Force-velocity relation + unbinding rate.

OUTPUT AND ANALYSIS

Kinesin + Dynein

Polarity sorting observed

Kinesin-driven antiparallel MT sliding and cortical dynein-mediated plus-end capture produce robust **microtubule polarity sorting**. MTs adopt a uniform plus-end-out orientation within protrusions, recapitulating the axon-like morphology characteristic of polarised neuronal processes. Process extension is spatially gated by the protrusion threshold, yielding stable, well-defined neurite-like structures.



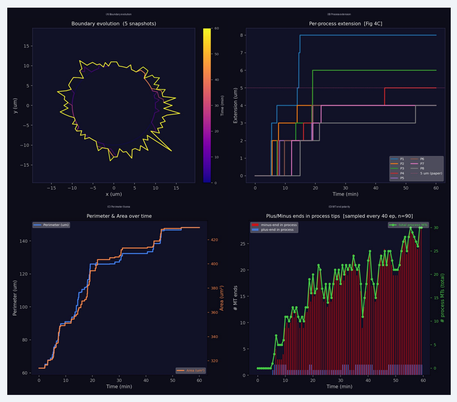
$N_{\mathrm{MT}} = 150$, $N_{\mathrm{kin}} = 200$, $\rho_{\mathrm{dyn}} = 0.25\,\mu\mathrm{m}^{-1}$, 1 h

Kinesin only

Growth without polarity sorting

In the absence of cortical dynein, kinesin-driven MT sliding generates sufficient force to deform the membrane boundary and drive morphological extension. However, without the orienting influence of dynein-mediated cortical capture, **polarity sorting is hindered** — MTs retain a predominantly minus-end-out disposition within protrusions. The cytoskeletal architecture is thus inverted relative to the physiological state.

**MTs initialised away from boundary in all simulations (Radius=7)



$N_{\mathrm{MT}} = 150$, $N_{\mathrm{kin}} = 200$, $\rho_{\mathrm{dyn}} = 0$, 1 h

Cortical dynein is exclusively membrane-anchored, exerting predominantly tangential forces on centrally located MTs — explaining the absence of protrusion in dynein-only configurations and highlighting kinesin's indispensable role in generating the initial outward displacement required for cortical engagement.

SUMMARY

- Implemented the Oelz 2018 actin–MT–cortical force model as an iterative Cytosim simulation with Python's management.
- Boundary polygon deformed each epoch via a five-stage protrusion rule; MT and motor positions carried forward.
- Radial force filtering and post-protrusion smoothing eliminate artefacts and produce clean protrusion profiles.
- Employs membrane tension & differential protrusion threshold to limit uncontrolled protrusion growth.

FUTURE DIRECTIONS

1. Local activation – global inhibition

Couple epoch simulation with a Turing-class LAGI framework to allow cortical dynein density to self-organise into discrete high-activity domains without an externally imposed cue.

2. Selective dynein recruitment

Incorporate compartment-specific dynein enrichment rules (AIS barrier, BICD2/Rab6 adaptors) to test whether differential motor density alone produces axon–dendrite cytoskeletal asymmetry.

3. Stochastic motor kinetics

Implement force-dependent catch-bond unbinding for dynein to simulate realistic motor turnover across simulations.

4. 3D boundary deformation

Extend planar polygon to a triangulated membrane mesh, capturing out-of-plane protrusion dynamics and membrane curvature coupling.

ACKNOWLEDGEMENTS

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- Cytosim engine: F. Nédélec & D. Foethke
- Model parameters: Oelz, del Castillo, Gelfand & Mogilner (2018)

REFERENCES

[1] Oelz et al. (2018). Microtubule dynamics, kinesin-1 sliding, and dynein action drive growth cone turning. *Biophys. J.* **115**, 1614–1624.
[2] Nédélec & Foethke (2007). Collective Langevin dynamics of flexible cytoskeletal fibers. *New J. Phys.* **9**, 427.
[3] Cytosim: `https://gitlab.com/f-nedelec/cytosim`
[4] Lu, W., Lakonishok, M., Liu, R., Bharat, V. & Gelfand, V. I. (2013). Initial neurite outgrowth in *Drosophila* neurons is driven by kinesin-powered microtubule sliding. *Curr. Biol.* **23**, 1018–1023.
[5] Bradke, F. & Dotti, C. G. (1999). The role of local actin instability in axon formation. *Science* **283**, 1931–1934.
[6] Witte, H., Neukirchen, D. & Bradke, F. (2008). Microtubule stabilization specifies initial neuronal polarization. *J. Cell Biol.* **180**, 619–632.
[7] Baas, P. W., Deitch, J. S., Black, M. M. & Banker, G. A. (1988). Polarity orientation of microtubules in hippocampal neurons: uniformity in the axon and nonuniformity in the dendrite. *Proc. Natl Acad. Sci. USA* **85**, 8335–8339.
[8] Kapitein, L. C. & Hoogenraad, C. C. (2011). Which way to go? Cytoskeletal organization and polarized transport in neurons. *Mol. Cell. Neurosci.* **46**, 9–20.